Team Number:	101
Project Name:	Team 101
Team Member Names:	Gary Hoffland, Caleb Ross, Dak
Version:	1

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage

All Major Components	**Component Name**	**Part Number**
PIC Microcontroller	PIC Microcontroller	PIC18F57Q43
	Quad Op-Amp	LM324N
	Reflective Optical Sensor	OPB732
	Piezo Buzzer	PT-1407
	Debug LED (red)	Standard LED
+5V Power Rail	**Component Name**	**Part Number**
	PIC Microcontroller	PIC18F57Q43
	Quad Op-Amp	LM324N
	Reflective Optical Sensor	OPB732
	Piezo Buzzer	PT-1407
	Debug LED (red)	Standard LED

c2. Regulator or Source Cl	+5V Regulator	LM7805

External Power Source 1	**Component Name**	**Part Number**
Power Source 1 Selection	9V PP3 Alkaline Battery	6LR61 / 9V Heavy Duty
Power Rails Connected	+5V Regulator	LM7805
to External Power		
Source 1		
		Total R

	Component Name	**Part Number**
	9V Alkaline Battery	6LR61

Power Budget

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e regulators, resistors, capacitors, or passive elements

SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)
+1.8V - 5.5V	1	30.0	30.0
+3.0V - 32V	1	3.0	3.0
+5.0V	1	17.3	17.3
+3.0V - 15V	1	15.0	15.0
+1.4V - 2.0V	1	9.1	9.1
SupplyVoltageRange	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**
+1.8V - 5.5V	1	30.0	30.0
+3.0V - 32V	1	3.0	3.0
+5.0V	1	17.3	17.3
+3.0V - 15V	1	15.0	15.0
+1.4V - 2.0V	1	9.1	9.1
		Subtotal	**74.4**
		Safety Margin	25%
Total Current Required on +5V Rail			**93.0**

+7V - 35V	1	1000	1000
Total Remaining Current Available on +5V Rail			**907.0**

SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)
9V DC	9V	500	500
+7V - 35V	1	1000	93
			0
			0
emaining Current Available on External Power Source 1			**407.0**

SupplyVoltageRange		Capacity(mAh)	RequiredByRegulators
9V DC		550	93
		Battery Life	**5.9**

Unit
mA
mA
mA
mA
mA
Unit
mA
mA
mA
mA
mA
mA
mA

mA
mA

Unit
mA
mA
mA
mA
mA

mA
hours